AERIS – Part 1 Progress Report

CSE518 Artificial Intelligence – Monsoon 2026

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1 Objective

Part 1 of AERIS implements the deterministic routing stage of the medical-supply delivery system. The road network is treated as fully observable: each road is either known to be open or known to be blocked. The system combines a small reactive control layer with graph search. Reactive rules handle immediate feasibility checks, while the deliberative layer uses BFS, UCS and A* to select a route from the depot to the requested destination.

The aim of this stage is to study A* in detail, especially the effect of different heuristic functions on search effort. Probabilistic road-state estimation and sequential decision making are kept for Part 2.

2 Search Problem Formulation

The deterministic routing problem is represented as follows.

Element	**Definition in AERIS Part 1**
State	Current node / vehicle position in the road network
Initial state	Depot
Actions	Traverse an adjacent road that is known to be open
Transition model	Move from the current node to the selected adjacent node
Goal test	Current node is the selected hospital, relief camp or demand point
Path cost	Sum of deterministic edge traversal costs

For A*, the evaluation function is

$$f(n) = g(n) + h(n),$$

where $g(n)$ is the accumulated path cost from the depot and $h(n)$ estimates the remaining cost to the goal. The application displays the expanded node, frontier, closed set and the corresponding $g(n)$, $h(n)$ and $f(n)$ values during the search.

3 Reactive Control

The reactive part of the system is intentionally simple in Part 1. It applies the following rules before or during route execution:

- a road marked as blocked is never used by the planner;
- if a known road state changes, the current route is invalidated and search is performed again;
- transport-mode feasibility is checked before a route is accepted;

- UAV-supporting legs retain a minimum battery reserve where applicable.

The important point in Part 1 is that these conditions are known. No probability distribution is used for road availability at this stage.

4 A* Heuristics

Four heuristics are implemented and compared.

Heuristic	**Definition**	**Role**
H1: Zero	$h_1(n) = 0$	UCS-equivalent baseline
H2: Geometric lower bound	Straight-line distance multiplied by the minimum observed cost per unit distance	Low-cost geometric estimate
H3: Relaxed network	Shortest-path cost in a lower-cost relaxed version of the road network	Relaxed-problem estimate
H4: Combined	$h_4(n) = \max(h_2(n), h_3(n))$	Tests heuristic dominance

For H3, each traversable edge in the relaxed graph uses

$$c_{\text{relaxed}}(e) = \min\big(c_{\text{original}}(e),\ d_{\text{geom}}(e)\big).$$

Since the relaxed edge cost is never larger than the original edge cost, the relaxed shortest-path value provides a lower bound on the original remaining path cost.

5 Comparison Results

The four heuristics were tested on three deterministic sample maps. The comparison records path cost, number of expanded nodes, runtime, admissibility and consistency. The mean results are shown below.

Heuristic	**Mean cost**	**Mean expanded**	**Admissible**	**Consistent**
H1: Zero	7.00	7.00	Yes	Yes
H2: Geometric	7.00	6.33	Yes	Yes
H3: Relaxed network	7.00	5.00	Yes	Yes
H4: $\max(H2, H3)$	7.00	5.00	Yes	Yes

All four heuristics produced the same optimal mean path cost on the bundled maps. H1 expanded the most nodes because it has no goal-directed information. H2 reduced the search slightly. H3 and H4 required fewer node expansions in these examples while remaining admissible and consistent. The purpose of the comparison is therefore not only to measure runtime, but also to observe how a more informed heuristic changes the amount of search performed.

6 External Map Input

The application is not restricted to one built-in graph. It can load an external road network in either JSON format or from CSV files containing nodes and edges, with an optional request file.

The importer checks for a Depot node, valid coordinates, positive costs for open roads and at least one valid delivery destination. Three JSON maps and CSV templates are included with the project.

7 Testing and Current Status

The final Part 1 implementation was tested after integrating the external-map and heuristic-comparison changes. The focused integration suite passed 150 of 150 checks, and the randomized regression suite passed 3423 of 3423 checks. The tests covered blocked-road handling, unreachable destinations, agreement between A* and UCS on optimal path cost, H1–H4 admissibility and consistency, external-map validation, and propagation of road edits across the search and comparison modules.

The following Part 1 deliverables are complete:

- Streamlit application for deterministic routing and A* visualization;
- BFS, UCS and A* implementations;
- four A* heuristics with admissibility and consistency checks;
- external JSON/CSV road-network input;
- heuristic-comparison notebook and result file;
- this progress report.

8 Next Stage

Part 2 will reuse the Part 1 route planner but will no longer assume that road state is perfectly known. The next stage will add Bayesian road-accessibility estimates, HMM-based temporal updates, risk-aware A* costs and an MDP for high-level sequential decisions. These extensions are kept separate from Part 1 so that the deterministic search component can be evaluated independently.